

RESEARCH ARTICLE



Real-world treatment patterns and attrition for non-driver mutation metastatic non-small cell lung cancer in the US

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ABSTRACT

Aims: Programmed cell death protein-1/ligand-1 (PD-1/PD-L1) inhibitors (PD-[L]1) are standard of care for patients with metastatic non-small cell lung cancer (mNSCLC). This study examined real-world treatment patterns and attrition by lines of therapy (LOTs) among patients with non-driver mutation mNSCLC.

Patients & methods: A retrospective study of adult patients with mNSCLC (2015–2022) who received ≥ 1 systemic treatment in COTA's United States multicenter NSCLC database was conducted. Treatment patterns, duration, and outcomes were summarized descriptively. PD-(L)1 utilization was stratified by PD-L1 expression levels ($<1\%$, $1\%–49\%$, $\geq 50\%$).

Results: Among the 2,107 eligible patients, PD-(L)1-based therapy was the most common frontline therapy (55.8%); of these, 60.5% received PD-(L)1/platinum combination therapy. Among PD-(L)1 users, frontline PD-(L)1 monotherapy was most frequently utilized in patients with PD-L1 expression $\geq 50\%$ (66.2%). Utilization of frontline platinum chemotherapy without PD-(L)1 decreased from 76.8% (2015) to $<15\%$ (2019–2022). Mortality across LOTs 1–4 was 37.4%–42.2% and attrition was approximately 60% for each LOT. The overall median duration of LOT1 was 5.4 months. A decreasing trend in the treatment and LOT duration of subsequent LOTs was observed.

Conclusions: Despite incorporating PD-(L)1-based therapies for frontline mNSCLC, mortality and attrition during LOT1 remained high and therapy duration was short, reflecting challenges in managing mNSCLC.

PLAIN LANGUAGE SUMMARY

People with advanced non-small cell lung cancer (NSCLC) are often treated with immunotherapies. These are drugs that help the body's immune system find and destroy cancer cells. One common drug type targets the programmed cell death protein-1/ligand-1 (PD-1/PD-L1; PD-[L]1) pathway, which cancer cells use to hide from the immune system. This study looked at how these treatments are used in real life among patients in the United States whose lung cancer did not have any specific genetic changes. Researchers used information from COTA's NSCLC database, which includes treatment data from clinics/hospitals across the country. They studied patients who received at least one systemic immunotherapy or chemotherapy between 2015 (when immunotherapy was first approved) and 2022. Out of 2,107 patients included in the study, doctors' prescribing patterns changed over time. Before immunotherapies became common, most patients were treated first with platinum-based chemotherapy. After immunotherapies were introduced, more patients began receiving PD-(L)1-based drugs as their first treatment, especially if their cancer cells showed high levels ($\geq 50\%$) of PD-L1. However, even with this shift to immunotherapy, treatment results were not ideal. Many patients stopped treatment early, often because the cancer continued to grow or due to death. Treatment durations were generally short, and most patients did not receive additional therapies after their first one. Overall, the study shows that while immunotherapies have changed how doctors treat advanced NSCLC, the outcomes for many patients remain poor. Better treatment strategies are still needed to improve patient survival and quality of life.

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1. Introduction

Non-small cell lung cancer (NSCLC), which accounts for ~85% of lung cancer cases globally as well as in the United States (US) [1–3], is mostly diagnosed at an advanced stage (Stage IIIB/C

and metastatic NSCLC [mNSCLC; i.e., Stage IV]) [2,4], making effective treatment options vital for patient survival and quality of life. Over the past decade, the treatment landscape for non-driver mutation advanced NSCLC (i.e., those not involving

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Article highlights

- Immunotherapies, including programmed cell death protein-1/ligand-1 (PD-1/PD-L1) inhibitors (PD-[L]1), are considered standard of care in patients with non-driver mutation metastatic non-small cell lung cancer (mNSCLC).
- This retrospective, real-world study in patients with non-driver mutation mNSCLC in the US demonstrated a shift in frontline treatment patterns from platinum chemotherapy toward PD-(L)1-based therapies between 2015 and 2022.
- Overall, attrition and mortality rates remained high, treatment durations were short, and most patients did not receive subsequent treatment following each line of therapy.
- Among PD-(L)1 users with high tumor PD-L1 expression ($\geq 50\%$), most received frontline PD-(L)1 monotherapy (66.2%), and the rest received PD-(L)1/platinum combination (Supplementary Table S7).
- Study findings underscore significant challenges in current treatment strategies and the need for more effective treatments (or sequence of treatments) for optimizing outcomes in patients with advanced disease.

genetic alterations in epidermal growth factor receptor [EGFR] or anaplastic lymphoma kinase [ALK]) has shifted from standard-of-care (SoC) chemotherapy to immunotherapy, either as a monotherapy or a combination therapy [5–8]. Recent advancements in immune checkpoint inhibitors (ICIs) have revolutionized the treatment paradigm and survival outcomes in patients with non-driver mutation advanced NSCLC [9–11]. ICIs, such as inhibitors of the programmed cell death protein-1/ligand-1 (PD-1/PD-L1) pathway, have emerged as a cornerstone of care as a first- or second-line therapy in the management of non-driver mutation advanced NSCLC [12–14], leading to prolonged survival in a subset of patients [15]. Moreover, the PD-L1 expression levels on tumors have been reported to correlate with the response to ICIs [14].

In patients with advanced NSCLC, irrespective of histology findings, clinical guidelines recommend testing tumor PD-L1 expression levels before initiating treatment to determine whether the treatment plan should include ICIs alone or in combination with chemotherapy [9]. The guidelines further suggest that the SoC in patients with non-driver mutation advanced NSCLC should be systemic therapies that include platinum-based chemotherapy and ICIs, unless the patient has contraindications to ICIs [16–18]. As part of frontline NSCLC management, PD-1/PD-L1 (PD-[L]1) inhibitors have been approved for treating advanced NSCLC either as a monotherapy [5,19], in combination with chemotherapy [20–22], or in combination with anti-cytotoxic T lymphocyte antigen-4 (CTLA-4) agents [8,15]. Nevertheless, primary- and secondary-resistance and progression after discontinuing immunotherapy are frequently observed in clinical practice [23,24].

While immunotherapies have evolved rapidly in recent years, there still exists a need to comprehensively explore how these recent approvals have translated into real-world treatment patterns and patient outcomes. While previous studies have examined treatment patterns with immunotherapies, our study describes the treatment patterns and attrition by line of therapy (LOT) among patients with non-driver mutation mNSCLC from the start of the

immunotherapy era in 2015 through 2022, capturing a period of newer approvals and evolving treatment patterns. Furthermore, the analysis includes outcomes stratified by PD-L1 expression levels, providing a comprehensive understanding of immunotherapy utilization in the routine clinical care settings.

2. Methods

2.1. Data source and study design

This was a retrospective cohort study utilizing COTA's US multicenter NSCLC Vantage Real-World Evidence database (Cota Inc.) [25]. The COTA database is a curated oncology-focused dataset consisting of longitudinal, Health Insurance Portability and Accountability Act 1996 (HIPAA)-compliant data on patient diagnoses, clinical management, tumor characteristics, and treatment outcomes. De-identified data from electronic medical records (EMR) of patients are obtained from an even mix of academic and community institutions across the US. In addition to typically available structured EMR data, curated data on LOTs and other tumor characteristics are also included.

To determine outcomes such as clinical progression and mortality, COTA utilizes structured and unstructured data from the EMRs of partnered healthcare provider sites and commercially available obituary data, including the Social Security Administration Death Master File. Based on available information, clinically trained abstractors review and confirm the data to ensure that the information reflects actual clinical progression or death. Regarding mortality, COTA's mortality algorithm showed strong agreement (88%) with the National Death Index [26].

The study design included a patient identification period and a follow-up period. The patient identification period was between 1 January 2015 and 31 December 2022. The follow-up period spanned from 1 January 2015 to 1 May 2023 (end of data cut), which allowed at least four months of follow-up for all eligible patients. The index date was defined as the date of the initial diagnosis of mNSCLC for eligible patients. For patients initially diagnosed with earlier stages of NSCLC that subsequently progressed to metastatic disease, the index date was defined as the date of distant progression. All eligible patients were followed from the index date until the last date of documented contact or treatment, date of death, or the end of data cut (1 May 2023), whichever occurred first.

2.2. Study population

Patients aged ≥ 18 years, newly diagnosed with mNSCLC (Stage IV disease) between 1 January 2015 and 31 December 2022, with at least one documented treatment for their metastatic disease initiated within 30 days before and up to 120 days after the mNSCLC diagnosis, were eligible for the study. Patients who had a documented positive test result for EGFR or ALK mutations, received tyrosine kinase inhibitors (TKIs) at any time during the patient treatment journey, or those with neuroendocrine carcinoma histologies were

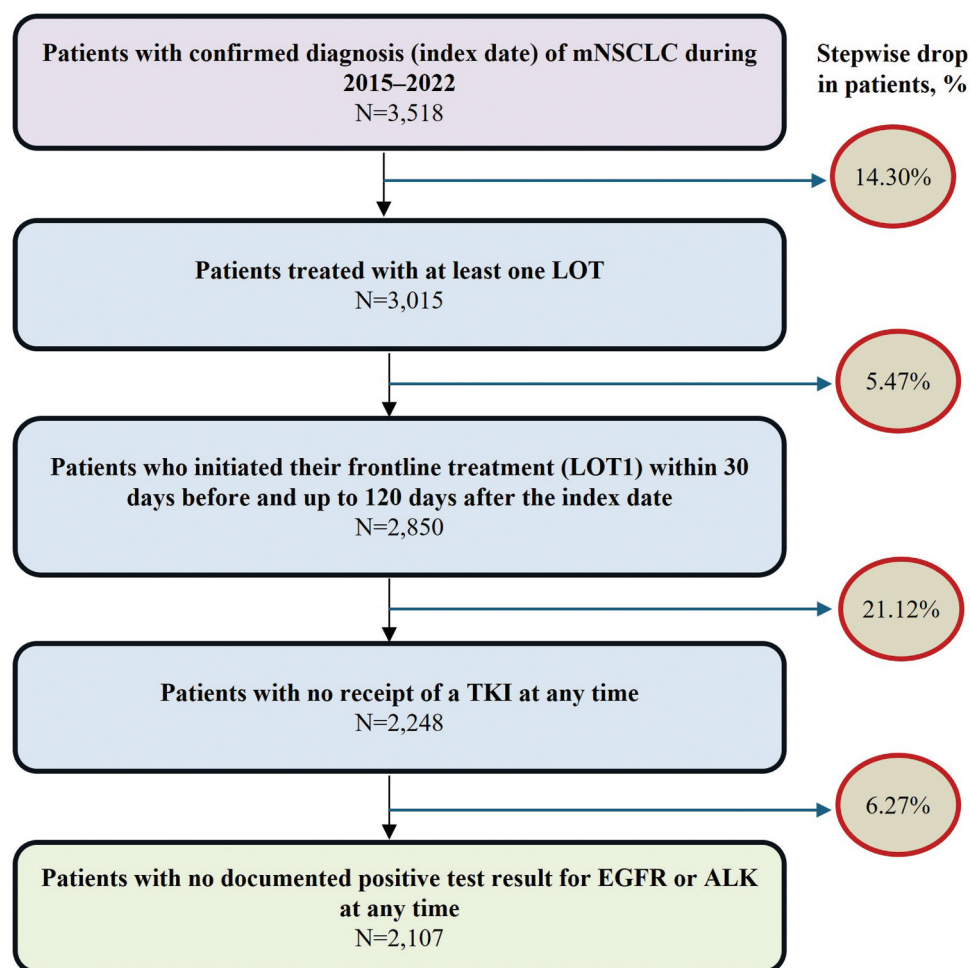


Figure 1. CONSORT flow diagram.

Abbreviations: ALK, anaplastic lymphoma kinase; CONSORT, Consolidated Standards of Reporting Trial; EGFR, epidermal growth factor receptor; LOT, line of therapy; mNSCLC, metastatic non-small cell lung cancer; TKI, tyrosine kinase inhibitor.

excluded. Of the 3,518 mNSCLC patients identified in the COTA database, 2,107 patients were eligible for analysis (Figure 1).

2.3. Statistical analyses

Baseline and disease characteristics in the frontline setting were summarized for each treatment group using continuous or categorical summary measures, as appropriate. The parameters captured included age, sex, race, histology, tobacco history, PD-L1 expression levels, the Eastern Cooperative Oncology Group (ECOG) score, presence of liver, brain, adrenal, or lymph node metastases at any time after initial diagnosis, and the type of clinical setting.

Anti-neoplastic drugs were classified into the following drug classes of interest: PD-(L)1, CTLA-4, platinum-based chemotherapy, non-platinum chemotherapy, unknown therapy, and other drugs (that do not fall into any of the above classes) – further details are provided in Supplementary Table S1. A comprehensive assessment of treatment patterns, duration, and outcomes among the study population was conducted and summarized using descriptive statistics. Utilization of different treatments in the frontline setting

by year of diagnosis was visualized using a bar chart. Frequency distributions for the different treatment groups as patients undergo each LOT up to the maximum number of LOTs available were captured to analyze the treatment patterns. Treatment sequences for the first four LOTs were illustrated using a Sankey diagram. The attrition counts and proportion of patients lost (including frequencies of death and no subsequent treatment or follow-up) with each LOT were reported.

The duration of treatment (active administration period) was calculated for each LOT, as was the overall LOT duration. Treatment duration reflects time spent on treatment, inclusive of maintenance therapy, within a specific LOT, and was defined as the time between the start date of a LOT and the last date of drug administration in that LOT. Duration of the LOT reflects time spent on the entire LOT, inclusive of treatment duration and any treatment-free interval before the subsequent LOT. Hence, the overall LOT duration was defined as the time between the start date of a LOT and one day before initiation of the subsequent LOT, last provider contact date, or death, whichever came first. Median and interquartile range (IQR) were used to summarize treatment and overall LOT duration.

Time to next treatment (TTNT) among patients initiating frontline PD-(L)1 treatment was defined as the interval from initiation of frontline systemic therapy to either the start of a subsequent line of therapy or death from any cause during frontline therapy (i.e., no second-line), whichever occurred first. Patients were censored at the last known follow-up if neither of the events occurred. Cox proportional hazards models and multivariate logistic regression analysis were used to estimate the association between baseline characteristics and 1) frontline TTNT or 2) frontline mortality. An indicator for PD-(L)1 monotherapy (referent) vs. PD-(L)1 combined with platinum chemotherapy was also included. Hazard ratios, odds ratios, and 95% confidence intervals (CIs) were reported.

Additional analyses were conducted on subsets of patients treated with PD-(L)1 inhibitors (either as monotherapy or as combination therapy) and further stratified by PD-L1 expression level (<1%, 1%–49%, and ≥50%). We utilized PD-L1 expression based on the tumor proportion score, as provided in the COTA database. SAS Version 9.4 (SAS Institute Inc., USA) was used to build the analytic dataset, and R version 4.3.2/ RStudio 2021.09.0 was used to perform all statistical analyses.

3. Results

3.1. Baseline demographics and clinical characteristics

At baseline, the mean (SD) age of patients was 67.9 (10.2) years, with the vast majority of patients being ≥65 years of age (63.3%) (Table 1). Patients were mostly White (74.1%), with slightly more males (53.8%) than females, and received

a comparable mix of cancer treatments in the academic (47.2%) and community (52.8%) settings (Supplementary Table S2). PD-L1 expression level was known for over half of the patients. A higher percentage of individuals aged over 75 years (39.2%), with an ECOG score of ≥2 (19.8%) and PD-L1 expression ≥50% (59.8%) was observed in patients receiving frontline PD-(L)1 monotherapy compared to those receiving frontline PD-(L)1 therapy in combination with platinum therapy. The frequency of brain metastases (36.5%) appeared higher among patients receiving frontline PD-(L)1 and platinum combination therapy.

3.2. Utilization of frontline treatment over time

The utilization of frontline platinum chemotherapy without PD-(L)1 decreased from 76.8% in 2015 to less than 15% during 2019–2022, paralleling an increase in the use of PD-(L)1 and platinum combination therapy during the same period (Figure 2). The utilization of frontline

PD-(L)1 monotherapy was the highest in 2018 (26.6%), following which there was a steady use (~20%) from 2019 to 2022. Between 2020 and 2022, the use of PD-(L)1 in combination with CTLA-4 almost doubled, albeit comprising only a small fraction of frontline mNSCLC treatment.

3.3. Treatment patterns

The mean (SD) follow-up time for study patients was 18.8 (1.3) months. Among the 2,107 eligible patients, PD-(L)1-based

Table 1. Baseline demographics and clinical characteristics.

Variables	Overall	Frontline PD-(L)1 monotherapy	Frontline PD-(L)1/platinum combination therapy
All, n (%)	2,107 (100.0)	393 (100.0)	710 (100.0)
Age, mean (SD)	67.9 (10.2)	71.7 (10.0)	66.9 (9.9)
Age category, n (%)			
<65 years	749 (35.5)	88 (22.4)	278 (39.2)
65–75 years	820 (38.9)	139 (35.4)	290 (40.8)
>75 years	515 (24.4)	154 (39.2)	142 (20.0)
Unknown	23 (1.1)	12 (3.1)	0 (0.0)
Histology, n (%)			
Non-squamous	1,356 (64.4)	238 (60.6)	476 (67.0)
Squamous	419 (19.9)	102 (26.0)	118 (16.6)
Not otherwise specified	332 (15.8)	53 (13.5)	116 (16.3)
Tobacco history, n (%)			
Yes	1,827 (86.7)	351 (89.3)	616 (86.8)
No	188 (8.9)	33 (8.4)	58 (8.2)
Unknown	92 (4.4)	9 (2.3)	36 (5.1)
ECOG score, n (%)			
0	398 (18.9)	80 (20.4)	141 (19.9)
1	719 (34.1)	140 (35.6)	243 (34.2)
≥2	340 (16.1)	78 (19.8)	81 (11.4)
Unknown	650 (30.8)	95 (24.2)	245 (34.5)
Metastases, n (%)			
Liver	505 (24.0)	82 (20.9)	167 (23.5)
Brain	715 (33.9)	115 (29.3)	259 (36.5)
Adrenal	436 (20.7)	82 (20.9)	146 (20.6)
Lymph node	359 (17.0)	60 (15.3)	114 (16.1)
PD-L1 expression level, n (%)			
<1%	265 (12.6)	11 (2.8)	128 (18.0)
1%–49%	405 (19.2)	47 (12.0)	202 (28.5)
≥50%	458 (21.7)	235 (59.8)	120 (16.9)
Unknown	979 (46.5)	100 (25.4)	260 (36.6)

Abbreviations: ECOG, Eastern Cooperative Oncology Group; PD-1, programmed cell death protein 1; PD-L1, programmed cell death ligand 1; PD-(L)1, anti-PD-1 and anti-PD-L1 drugs; SD, standard deviation.

Note: Treatment groups shown are a subset of frontline therapies of interest.

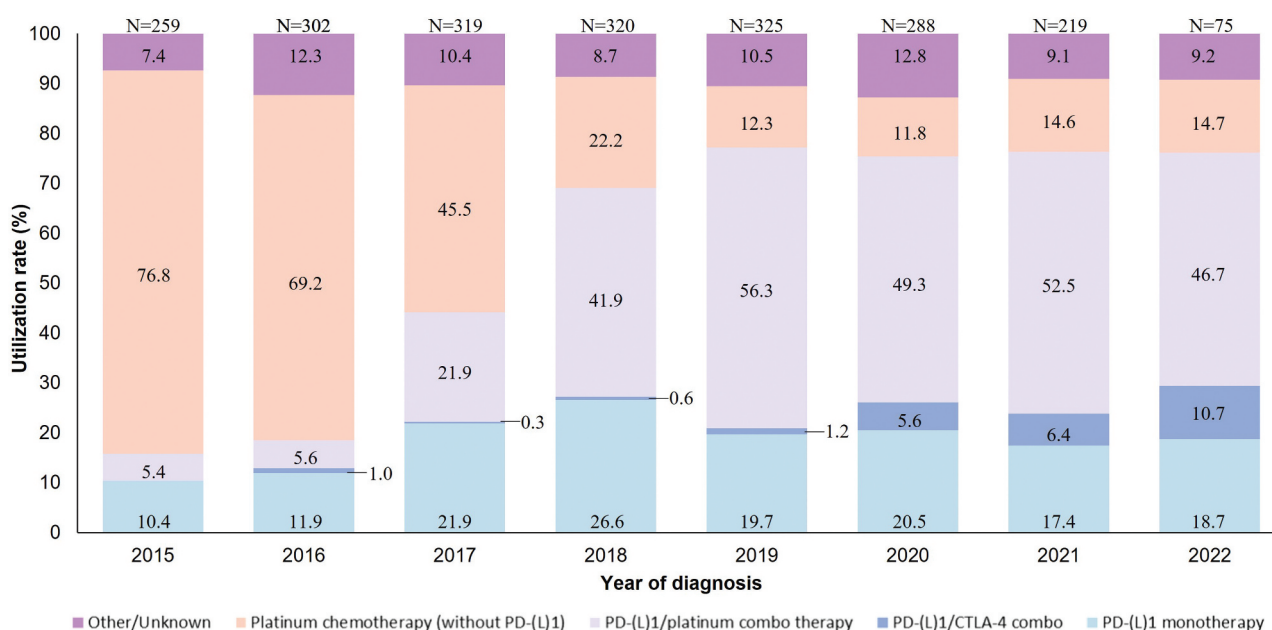


Figure 2. Utilization of frontline treatment by year of diagnosis.

Abbreviations: CTLA-4, cytotoxic T-lymphocyte – associated antigen 4; PD-1, programmed cell death protein 1; PD-L1, programmed cell death ligand 1; PD-(L)1, anti-PD-1 and anti-PD-L1 drugs.

therapy was the most common frontline therapy ($n = 1,174$, 55.8%), of which 710 (33.7%) were PD-(L)1/platinum combination and 393 (18.7%) were PD-(L)1 monotherapy users (Figure 3). In subsequent lines, PD-(L)1 monotherapy was most frequently used in the second-line, while non-PD-(L)1

and non-platinum chemotherapies were most frequently observed in third- and fourth-lines.

Pembrolizumab was the most frequently used PD-(L)1 agent as frontline therapy, both as monotherapy or in combination with platinum therapy, followed by nivolumab

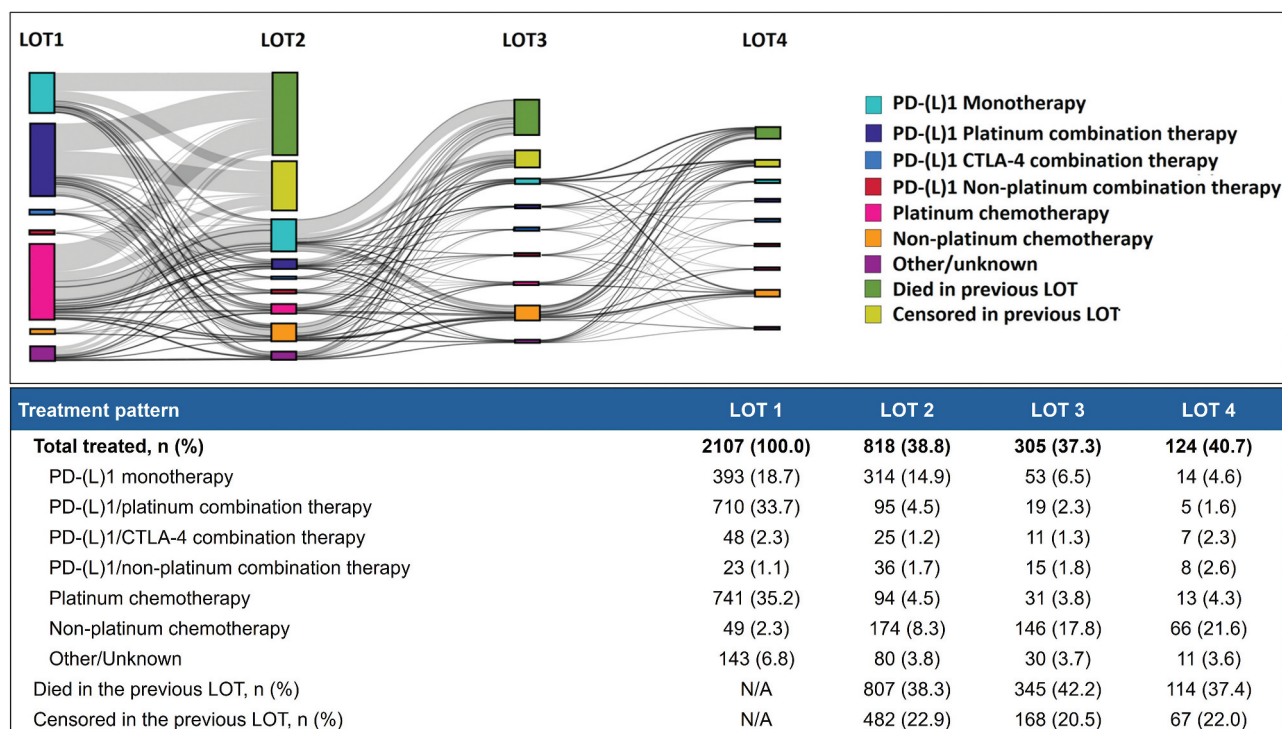


Figure 3. Sankey diagram of treatment patterns in patients with mNSCLC.

Abbreviations: CTLA-4, cytotoxic T-lymphocyte – associated antigen 4; LOT, line of therapy; mNSCLC, metastatic non-small cell lung cancer; PD-1, programmed cell death protein 1; PD-L1, programmed cell death ligand 1; PD-(L)1, anti-PD-1 and anti-PD-L1 drugs.

Percentages (%) were calculated among the total treated patients from the previous LOT.

(Supplementary Tables S3 and S4). Among patients with at least two LOTs, the most commonly used non-platinum chemotherapy as second-line therapy or beyond was docetaxel (40.4%), followed by pemetrexed (38.1%) (Supplementary Table S4). The most common second- and third-line therapies post-frontline PD-(L)1 monotherapy were platinum chemotherapy (25.4%) and non-platinum chemotherapy (55.6%), respectively, while non-platinum chemotherapy was the most common second- (44.1%) and third-line (54.2%) treatment post-frontline PD-(L)1/platinum combination therapy (Supplementary Table S5). The distribution of second-line treatments among those who had frontline PD-(L)1/platinum combination therapy is shown in Supplementary Table S6. When stratified by PD-L1 expression levels, among the patients who received frontline PD-(L)1 therapy, the use of PD-(L)1 monotherapy was infrequent at 7.9% and 18.9% in those with PD-L1 expression <1% and 1%–49%, respectively, and combination therapy was more common in these patients (Supplementary Table S7). Among patients with PD-(L)1 use during the first two LOTs, the vast majority (87.5%) had exposure to platinum chemotherapy. Specifically, in patients with PD-L1 expression ≥50%, nearly 1 in 4 patients (24.5%) did not receive platinum chemotherapy in the first two lines. Of note, among the 458 patients with PD-L1 expression ≥50%, although 355 patients received frontline PD-(L)1 therapy (either as monotherapy or combination therapy), 103 patients (22.5%) were not treated with any frontline PD-(L)1 therapy (Table 1).

3.4. Attrition and duration of therapy by LOT

Overall, approximately 60% attrition was observed for each LOT as patients progressed from LOT1 to LOT4 (Table 2). Death was the main reason for attrition, with a substantial proportion of patients dying during each LOT (37.4%–42.2% across LOTs 1–4). Specifically, during frontline treatment, 45.5% of PD-(L)1 monotherapy users and 37.9% of PD-(L)1/platinum combination therapy users died, and less than one-third of frontline PD-(L)1 users received subsequent treatment during the observed follow-up.

The overall median treatment duration and median LOT duration in LOT1 were 2.9 and 5.4 months, respectively, with the longest median duration observed among the frontline

PD-(L)1/platinum combination therapy users (4.2 and 7.3 months, respectively) (Table 2). A decreasing trend in the duration of each subsequent LOT was observed as patients advanced to later lines. Among those patients who received PD-(L)1/platinum chemotherapy in the frontline setting and initiated second-line therapy (N=220), only 42 (~19%) received platinum chemotherapy (with or without PD-[L]1) in the second-line setting (Supplementary Table S6). These patients appeared to have longer median (IQR) LOT duration on frontline PD-(L)1/platinum chemotherapy (11.0 [4.2–14.3] months) compared to those who did not receive any platinum chemotherapy (7.0 [4.0–10.8] months) in the second-line setting. Among patients treated with frontline PD-(L)1 monotherapy, although the counts of patients with lower PD-L1 expression levels were too small to yield meaningful conclusions, the results generally indicated a longer median treatment and LOT duration in patients with PD-L1 expression ≥50% (Table 3). A similar pattern was observed in patients treated with frontline PD-(L)1/platinum combination therapy, where, for instance, the median treatment duration in LOT1 for patients with PD-L1 expression ≥50% and <1% was 5.5 and 4.1 months, respectively. Similarly, median LOT1 duration for patients with PD-L1 expression ≥50% and <1% was 9.8 and 6.4 months, respectively.

Kaplan-Meier curves demonstrated longer TTNT among patients receiving PD-(L)1/platinum combination therapy (8.68 [95% CI: 7.63, 9.77] months vs. 6.41 [95% CI: 4.87, 7.37] months in patients receiving PD-[L]1 monotherapy). When further stratified by PD-L1 expression level, numerically longer TTNT was observed among patients with high PD-L1 expression level (≥50%), although the results were not statistically significant (Figure 4).

3.5. Predictors of frontline TTNT and mortality

The Cox proportional hazards models demonstrated that low PD-L1 expression (<1%), as well as brain, liver, or adrenal metastases were associated with a greater risk of

Table 2. Attrition and treatment duration by LOTs in patients with mNSCLC.

LOT	Number of patients	Attrition, %	Deaths, n (%)	No subsequent treatment during follow-up, n (%)	Subsequent treatment, n (%)	Treatment duration, months, median (IQR)	LOT duration, months, median (IQR)
Overall							
1	2,107	–	807 (38.3)	482 (22.9)	818 (38.8)	2.9 (1.0–6.6)	5.4 (2.4–11.5)
2	818	61.2	345 (42.2)	168 (20.5)	305 (37.3)	2.4 (1.0–5.6)	4.7 (2.3–9.6)
3	305	62.7	114 (37.4)	67 (22.0)	124 (40.6)	2.1 (0.8–5.1)	3.9 (2.1–7.9)
4	124	59.3	50 (40.3)	22 (17.8)	52 (41.9)	1.7 (0.7–4.0)	3.3 (1.9–6.9)
Frontline PD-(L)1 monotherapy users							
1	393	–	179 (45.5)	96 (24.4)	118 (30.0)	2.8 (0.7–7.0)	5.0 (1.7–14.0)
2	118	70.0	41 (34.7)	32 (27.1)	45 (38.1)	2.3 (1.1–6.4)	6.2 (2.8–10.8)
3	45	61.9	19 (42.2)	12 (26.7)	14 (31.1)	2.4 (1.4–4.8)	4.9 (2.9–9.5)
4	14	68.9	2 (14.3)	4 (28.6)	8 (57.1)	2.0 (1.1–5.7)	3.0 (2.1–8.2)
Frontline PD-(L)1/platinum combination therapy users							
1	710	–	269 (37.9)	221 (31.1)	220 (31.0)	4.2 (1.6–9.4)	7.3 (3.3–15.3)
2	220	69.0	101 (45.9)	47 (21.4)	72 (32.7)	2.1 (0.8–5.0)	4.1 (2.4–7.9)
3	72	67.3	29 (40.3)	18 (25.0)	25 (34.7)	1.4 (0.9–2.9)	2.9 (1.7–5.4)
4	25	65.3	16 (64.0)	2 (8.0)	7 (28.1)	1.0 (0.3–2.1)	2.6 (1.5–4.3)

Abbreviations: IQR, interquartile range; LOT, line of therapy; mNSCLC, metastatic non-small cell lung cancer; PD-1, programmed cell death protein 1; PD-L1, programmed cell death ligand 1; PD-(L)1, anti-PD-1 and anti-PD-L1 drugs.

Table 3. Attrition and treatment duration by LOT in patients with mNSCLC treated with frontline PD-(L)1 inhibitors, stratified by PD-L1 expression.

LOT	Number of patients	Attrition, %	Deaths, n (%)	No subsequent treatment during follow-up, n (%)	Subsequent treatment, n (%)	Treatment duration, months, median (IQR)	LOT duration, months, median (IQR)
Frontline PD-(L)1 monotherapy							
PD-L1 expression <1%							
1	11	–	5 (45.5)	2 (18.2)	4 (36.4)	1.6 (0.2–5.3)	2.1 (1.1–5.6)
2	4	63.7	2 (50.0)	0 (0.0)	2 (50.0)	1.7 (1.1–2.8)	2.9 (2.8–3.7)
3	2	50.0	1 (50.0)	0 (0.0)	1 (50.0)	4.7 (3.3–6.1)	6.9 (6.2–7.7)
4	1	50.0	0 (0.0)	0 (0.0)	1 (100.0)	1.0 (1.0–1.0)	1.5 (1.5–1.5)
PD-L1 expression 1%–49%							
1	47	–	25 (53.2)	11 (23.4)	11 (23.4)	2.8 (0.9–5.8)	4.5 (1.6–7.9)
2	11	76.6	5 (45.5)	3 (27.3)	3 (27.3)	1.6 (0.9–2.0)	3.0 (2.5–6.2)
3	3	72.8	0 (0.0)	1 (33.3)	2 (66.7)	6.1 (5.3–12.6)	22.4 (14.5–32.5)
4	2	33.3	1 (50.0)	0 (0.0)	1 (50.0)	2.0 (1.8–2.2)	3.6 (3.2–4.0)
PD-L1 expression ≥50%							
1	235	–	98 (41.7)	65 (27.7)	72 (30.6)	2.9 (0.7–7.8)	5.8 (1.7–15.2)
2	72	69.4	20 (27.8)	21 (29.2)	31 (43.1)	3.8 (1.4–6.9)	7.3 (3.4–13.6)
3	31	57.0	16 (51.6)	9 (29.0)	6 (19.4)	2.2 (0.8–3.2)	4.9 (2.3–9.1)
4	6	80.6	1 (16.7)	3 (50.0)	2 (33.3)	5.2 (3.1–5.5)	2.1 (1.8–5.6)
Frontline PD-(L)1/platinum combination therapy							
PD-L1 expression <1%							
1	128	–	51 (39.8)	28 (21.9)	49 (38.3)	4.1 (1.5–7.1)	6.4 (3.5–10.4)
2	49	61.7	28 (57.1)	6 (12.2)	15 (30.6)	2.1 (1.0–5.2)	4.7 (2.7–7.1)
3	15	69.3	5 (33.3)	7 (46.7)	3 (20.0)	1.2 (0.7–2.4)	2.1 (1.6–5.9)
4	3	80.0	3 (100.0)	0 (0.0)	0 (0.0)	2.0 (1.1–2.9)	4.3 (4.1–7.7)
PD-L1 expression 1%–49%							
1	202	–	75 (37.1)	70 (34.7)	57 (28.2)	3.5 (1.4–9.8)	7.5 (3.1–15.6)
2	57	71.8	22 (38.6)	16 (28.1)	19 (33.3)	2.3 (0.7–4.1)	4.5 (2.9–8.6)
3	19	66.7	8 (42.1)	4 (21.1)	7 (36.8)	1.7 (1.2–2.0)	2.9 (1.8–3.2)
4	7	63.2	2 (28.6)	2 (28.6)	3 (42.9)	1.1 (0.7–2.0)	2.4 (1.5–2.7)
PD-L1 expression ≥50%							
1	120	–	40 (33.3)	40 (33.3)	40 (33.3)	5.5 (1.4–12.1)	9.8 (5.0–23.7)
2	40	66.6	16 (40.0)	7 (17.5)	17 (42.5)	3.0 (1.2–7.5)	6.3 (2.8–9.1)
3	17	57.5	7 (41.2)	2 (11.8)	8 (47.1)	1.2 (0.8–5.3)	3.6 (1.8–6.4)
4	8	53.0	6 (75.0)	0 (0.0)	2 (25.0)	1.4 (0.9–3.1)	3.1 (1.6–4.6)

Abbreviations: IQR, interquartile range; LOT, line of therapy; mNSCLC, metastatic non-small cell lung cancer; PD-1, programmed cell death protein 1; PD-L1, programmed cell death ligand 1; PD-(L)1, anti-PD-1 and anti-PD-L1 drugs.

initiating second-line/death, whereas frontline PD-(L)1/platinum combination therapy, lower ECOG score, and history of tobacco use were associated with a lower risk of initiating second-line/death (Table 4). Similarly, multivariate logistic regression analysis demonstrated that younger age, lower ECOG score, later NSCLC diagnosis (2022), and having lymph node metastases was associated with a lower risk of frontline mortality (Supplementary Table S8).

4. Discussion

Findings from our retrospective cohort study in patients with non-driver mutation mNSCLC receiving at least one line of treatment suggest that PD-(L)1 therapy, either as a monotherapy or combined with chemotherapy, is the most common frontline therapy. Despite a considerable shift toward the use of immunotherapies, particularly PD-(L)1 inhibitors, the overall mortality during frontline treatment was nearly 40%. Furthermore, the study showed continued high attrition during each LOT and short duration of therapy, highlighting the need for effective treatment strategies across lines in these patients.

Most patients in our study were slightly older than those enrolled in NSCLC clinical trials [7,8,21,27]; however, demographic distributions were consistent with those of patients in other US-based studies of real-world mNSCLC patients receiving frontline therapies [28,29]. The adoption of

immunotherapies, including PD-(L)1 inhibitors, was similar across both academic and community settings, demonstrating their broad clinical acceptance and reflecting current real-world treatment practices. In addition, the findings highlight differences in treatment patterns based on PD-L1 expression, suggesting the need for tailored approaches to managing patients with non-driver mutation mNSCLC in routine clinical practice.

In our study, the utilization of frontline platinum chemotherapy without PD-(L)1 decreased substantially from 76.8% (2015) to <15% (2019–2022, each year). A similar decline was observed in a real-world Swedish National Lung Cancer Registry study in patients with advanced NSCLC within the same study period, from 72.2% (2017) to 25.1% (2020) [30]. The utilization of frontline PD-(L)1 monotherapy in the current study peaked at 26.6% in 2018 and decreased to 18.7% in 2022. However, its relatively steady use (~20% of frontline treatment) from 2019 to 2022 reflects the clinical practice guidelines and physician preferences for patients with high PD-L1 expression or those ineligible for combination therapies. Nevertheless, during the same period, the utilization of PD-(L)1/platinum combination therapy remained nearly twice as much as that of monotherapy. As such, treatment patterns in our study showed 55.8% of patients receiving PD-(L)1-based frontline therapy, amongst which the majority were combination therapies (66.5%, including chemotherapy or anti-CTLA-4 combination). These proportions were slightly higher

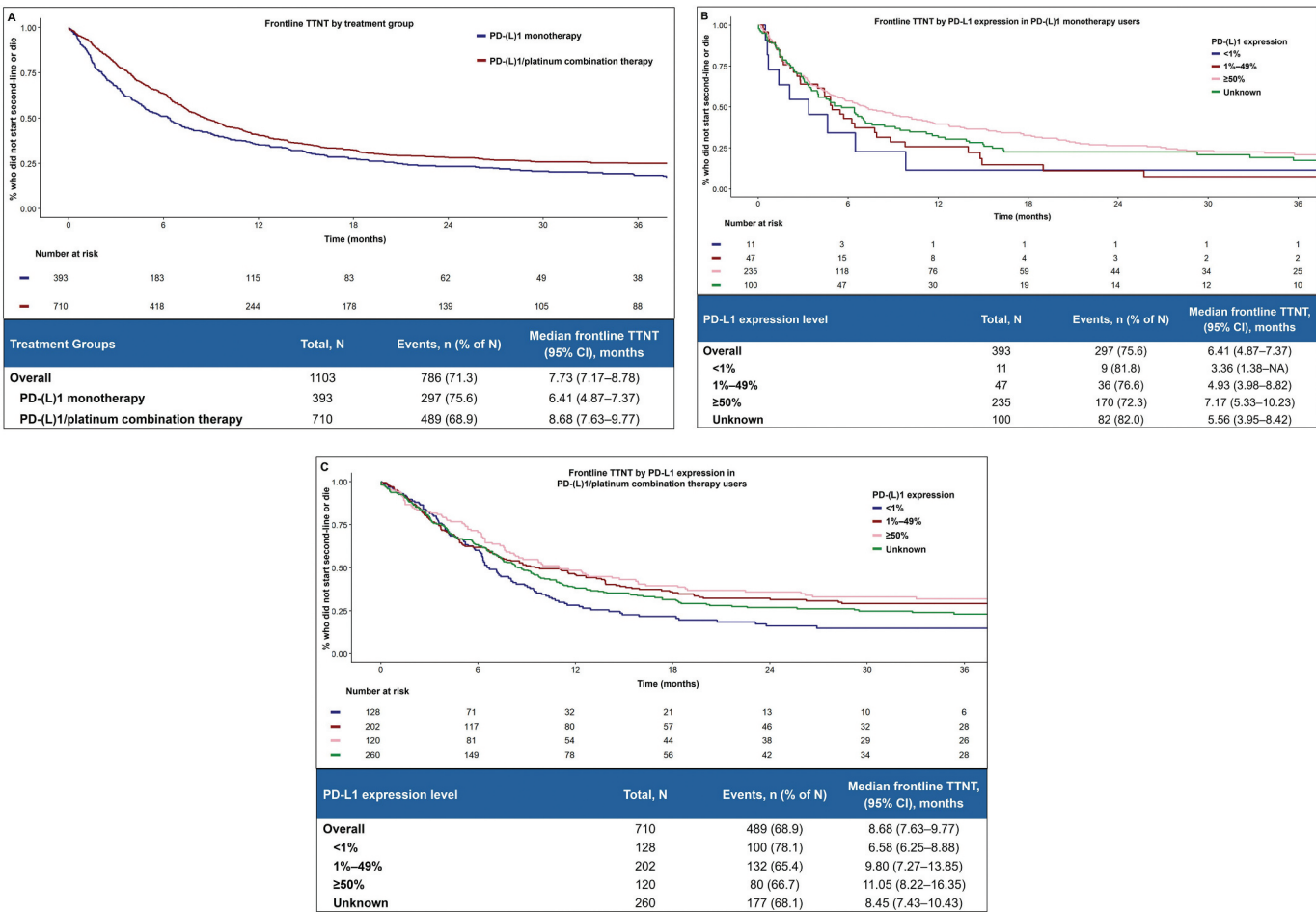


Figure 4. Kaplan – Meier curves for frontline TTNT (A) by treatment groups and by PD-L1 expression levels in (B) PD-(L)1 monotherapy users and (C) PD-(L)1/platinum combination therapy users.

Abbreviations: CI, confidence interval; PD-1, programmed cell death protein 1; PD-L1, programmed cell death ligand 1; PD-(L)1, anti-PD-1 and anti-PD-L1 drugs; TTNT, time to next treatment.

compared to another US-based retrospective study of adults with non-driver mutation mNSCLC who initiated frontline treatment between 1 August 2018 and 31 December 2019, in which ~55% of patients received a combination of immunotherapy and chemotherapy [28]. Overall, the shift observed in the utilization of frontline treatment over time in our study was consistent with findings from a systematic review on the topic [31] and aligns with the positive outcomes of clinical trials and the US regulatory approval of immunotherapy-chemotherapy combination regimens [32–35].

Despite the observed increase in the use of PD-(L)1-based therapies in our study, treating patients with NSCLC remains a challenge, as not all patients respond to immunotherapies (primary resistance), and some develop resistance soon after an initial response (acquired resistance) [23]. This was evident in a review of clinical trials evaluating ICIs in patients with NSCLC, which showed that the incidence of primary resistance to ICIs was the highest (41%–44%) in patients previously treated with chemotherapy and lowest (7%–11%) in those treated with immunotherapy plus chemotherapy as frontline therapy. With regards to secondary resistance, the incidence was 52%–57% in patients treated with ICIs as frontline therapy, and 32%–64% as second-line treatment (32%–64%) [23]. Irrespective of the

clinical guidelines recommending the usage of PD-(L)1-based regimens as frontline SoC for patients with advanced NSCLC [12,36], the utilization of frontline platinum chemotherapy was still close to 10% in recent years, as observed in our study. Furthermore, over a fifth of patients with high PD-L1 expression did not receive any frontline PD-(L)1 inhibitors. The nonuse of PD-(L)1-based regimens could likely be due to the current challenges associated with ICIs, which include contraindications to these therapies, their high cost, and limited access [37,38], although these parameters were not assessed, as it was not in the scope of the current study.

A high attrition was observed in our analysis of patients with non-driver mutation mNSCLC – over 60% per LOT. During frontline treatment, 20.5% of patients did not receive subsequent treatment, and the overall mortality was nearly 40%. Similar findings have been reported in several recent US-based real-world studies [28,39]. A real-world study that utilized another US-based database, Flatiron Health (>90% community-based practices), examined survival outcomes following frontline immunotherapy in patients with advanced NSCLC [40]. The analysis found that certain subgroups (i.e., older patients, those with poor performance status, lower PD-L1 expression, or brain metastases) were associated with shorter

Table 4. Predictors of frontline TTNT using Cox proportional hazards models.

Variables	n (%)	Frontline TTNT	
		Hazard Ratio (95% CI)	p-value
PD-(L)1 in LOT1			
PD-(L)1 monotherapy	393 (35.6)	Reference	
PD-(L)1/platinum combination therapy	710 (64.4)	0.75 (0.62, 0.90)	0.0019
Sex			
Male	593 (53.8)	Reference	
Female	509 (46.2)	0.91 (0.78, 1.05)	0.1974
Unknown	1 (0.1)	1.2 (0.16, 9.04)	0.8568
Age category, years			
<65	366 (33.2)	0.83 (0.69, 1.01)	0.0669
65–75	429 (38.9)	0.84 (0.70, 1.01)	0.0617
>75	296 (26.8)	Reference	
Unknown	12 (1.1)	4.16 (2.09, 8.27)	0.0001
Race			
White	804 (72.9)	Reference	
Black or African American	112 (10.2)	0.89 (0.70, 1.15)	0.3751
Asian	15 (1.4)	0.98 (0.51, 1.85)	0.9394
Other race	45 (4.1)	0.95 (0.67, 1.36)	0.7937
Unknown	127 (11.5)	0.98 (0.77, 1.24)	0.8422
PD-L1 expression level			
<1%	139 (12.6)	1.57 (1.22, 2.02)	0.0005
1%–49%	249 (22.6)	1.16 (0.93,1.44)	0.1857
≥50%	355 (32.2)	Reference	
Unknown	360 (32.6)	1.31 (1.08, 1.60)	0.0068
ECOG score			
0	221 (20.0)	0.48 (0.37, 0.61)	<0.0001
1	383 (34.7)	0.57 (0.46, 0.71)	<0.0001
≥2	159 (14.4)	Reference	
Unknown	340 (30.8)	0.56 (0.45, 0.70)	<0.0001
Histology			
Non-squamous	714 (64.7)	Reference	
Squamous	169 (15.3)	1.11 (0.92, 1.34)	0.2926
Not otherwise specified	220 (20.0)	1.15 (0.94, 1.41)	0.1623
Type of setting			
Academic	501 (45.4)	Reference	
Community	602 (54.6)	1.03 (0.88, 1.20)	0.6973
Tobacco history			
No	91 (8.3)	Reference	
Yes	967 (87.7)	0.70 (0.55, 0.90)	0.0053
Unknown	45 (4.1)	0.65 (0.43, 1.00)	0.0525
Brain metastases			
Without	729 (66.1)	Reference	
With	374 (33.9)	1.18 (1.01, 1.37)	0.0353
Adrenal metastases			
Without	875 (79.3)	Reference	
With	228 (20.7)	1.27 (1.07, 1.52)	0.0068
Liver metastases			
Without	854 (77.4)	Reference	
With	249 (22.6)	1.51 (1.28, 1.79)	<0.0001
Lymph node metastases			
Without	929 (84.2)	Reference	
With	174 (15.8)	0.88 (0.72, 1.07)	0.2089
Year of index date			
2015	41 (3.7)	Reference	
2016	53 (4.8)	1.25 (0.79, 1.99)	0.3412
2017	140 (12.7)	1.05 (0.70, 1.56)	0.826
2018	219 (19.9)	1.16 (0.78, 1.72)	0.4786
2019	247 (22.4)	1.13 (0.76, 1.67)	0.5392
2020	201 (18.2)	0.90 (0.60, 1.35)	0.6147
2021	153 (13.9)	1.07 (0.7, 1.62)	0.7668
2022	49 (4.4)	0.68 (0.37, 1.23)	0.1988

Abbreviations: ECOG, Eastern Cooperative Oncology Group; LOT, line of therapy; PD-1, programmed cell death protein 1; PD-L1, programmed cell death ligand 1; PD-(L)1, anti-PD-1 and anti-PD-L1 drugs; TTNT, time to next treatment. Note: Values highlighted in bold indicate significant risk factors.

median OS, consistent with the predictors of TTNT or death identified in the current study. Together with the current study's findings, high attrition and mortality and decreasingly shorter treatment durations in later lines underscore the rapid progression of the disease. Effective treatment strategies are needed to sustain therapeutic benefits and improve survival outcomes, not just in the US, but also in other regions such as

Europe and Asia [24]. Collectively, these data suggest a significant challenge in maintaining patient eligibility for subsequent therapies, highlighting the need to explore optimized sequencing of available therapies to maximize patient outcomes.

Pembrolizumab, a PD-1 inhibitor, is considered the frontline SoC treatment option for patients with advanced NSCLC and

PD-L1 expression $\geq 50\%$ without any contraindications to immunotherapy [35]. In our study, over 50% of patients with PD-L1 expression $\geq 50\%$ received frontline PD-(L)1 monotherapy, with pembrolizumab being the most frequently used frontline drug across all patients, consistent with real-world study reports [28]. This finding aligns with evidence from pivotal phase III clinical trials (KEYNOTE-024 and KEYNOTE-042), which show that the subset of patients with high PD-L1 expression may derive substantial benefit from PD-(L)1 monotherapy versus chemotherapy [19,41]. In one such phase III study conducted across 32 countries, patients with PD-L1 expression $\geq 50\%$ treated with pembrolizumab monotherapy had longer median overall survival compared to those with lower PD-L1 expression [41]. Similarly, across three real-world studies involving patients with advanced NSCLC, it was observed that frontline pembrolizumab monotherapy demonstrated overall survival benefits in patients with PD-L1 expression $\geq 50\%$ [42–44]. Despite the high utilization of PD-(L)1 monotherapy in the subset of frontline PD-(L)1 users with high PD-L1 expression in the current study, it was observed that one-third of such patients received frontline PD-(L)1/platinum combination therapy.

The median treatment and LOT duration in LOT1 appeared to be longer in patients treated with PD-(L)1 combination therapy than in those treated with PD-(L)1 monotherapy. Since this finding is consistent with other reports of real-world mNSCLC patients [28], there is support for novel combination treatments that may further improve patient outcomes and change the treatment landscape. Results were also suggestive of the importance of prolonged treatment and LOT duration in patients with higher PD-L1 expression levels, highlighting the need to develop better strategies for patients with lower PD-L1 expression to optimize treatment outcomes. Future research should monitor the evolving treatment landscape and evaluate potential impact on patient outcomes and attrition rates by novel immunotherapy approaches currently being investigated for solid tumors, including bispecific antibodies, lymphocyte activation gene-3 (LAG3), T cell immunoglobulin and mucin-domain containing 3 (TIM3), and T cell immunoglobulin and ITIM domain (TIGIT), and combination therapies with antibody – drug conjugates [45–47].

A strength of our study was the use of real-world data from the COTA US multi-center NSCLC database. The database contains an equal mix of patients in academic vs. community treatment settings [25], reflecting routine clinical practice across both settings in recent years. Our study provides comprehensive insights into the evolving landscape of advanced NSCLC treatment, highlighting changes in practice patterns in the immunotherapy era. However, certain limitations should be noted. The retrospective study design and reliance on EMRs that capture real-world data have the potential for missing or incomplete data capture. The study may have been subject to confounding, where the severity of the patient's condition or overall health could have affected both the treatment prescribed and the patient's prognostic outcome. Determining PD-L1 expression in advanced NSCLC is essential for guiding treatment decisions and determining eligibility for immune checkpoint inhibitors; hence, missing PD-L1 expression data represents a limitation of this real-world analysis. Furthermore, while our analysis focused on systemic therapy

treatment patterns, palliative or local ablative therapy is possible; however, it was outside the scope of this analysis. The study's descriptive estimates may be impacted by unmeasured confounding and missing data, as is commonly observed in retrospective epidemiological studies. Finally, since the study was not designed to evaluate patient outcomes, it was not possible to draw any conclusions regarding prognoses.

5. Conclusion

Although PD-(L)1-based therapies are increasingly used as frontline treatment for mNSCLC, their duration remains relatively short, and most patients do not proceed to subsequent LOTs. Among those who do, treatment duration progressively declines with each LOT, underscoring persistent challenges in disease management strategies and emphasizing the need for more sustained and effective therapeutic approaches in frontline and advanced settings. Future research should focus on identifying novel treatment approaches that can extend the duration of therapy, improve survival outcomes, and better manage disease progression in real-world clinical practice.

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Author contributions

Concept and design: AJS, CH, UC, QH, AH. Acquisition, analysis, or interpretation of data: All authors. Drafting of the manuscript: UC, CW, QH. Critical revision of the manuscript for important intellectual content: All authors. Statistical analysis: CH, UC, CW, TW, QH. Administrative, technical, or material support: UC, CW, TW, QH. Supervision: QH. Accountable for all aspects of the work: All authors.

Disclosure statement

Adam J Schoenfeld, Chen Hu, Ravi Rajaram, and Josephine Feliciano are consultants for Johnson & Johnson, USA. Urmila Chandran was an employee of Johnson & Johnson, USA at the time of study conduct and completion. Charlene Wong, Iftekhar Kalsekar, Tianyi Wang, Qing Huang, and Aisha Hasan are employees of Johnson & Johnson, USA. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

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Ethical declaration

COTA's de-identified real-world database was approved by the WCG Institutional Review Board for use in research (Study #1174746). The current study involved secondary analysis of a fully de-identified database; therefore, it was exempt from Institutional Review Board review, and informed consent was not required in accordance with 45 CFR 46 (US). The study was conducted in compliance with all applicable regulations and ethical guidelines.

Data availability statement

The data underlying this article were provided by COTA, Inc. and have sharing restrictions, as the licensed data is a proprietary asset of COTA, Inc. Summary-level data are provided throughout the manuscript and in the accompanying tables and figures. Any additional information or data that supports the findings of this study are available from the corresponding author upon reasonable request.

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